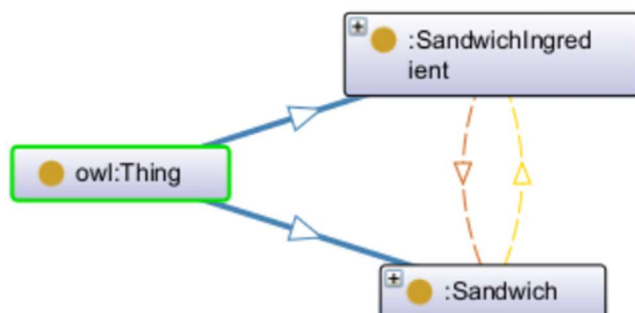


Exercise E4

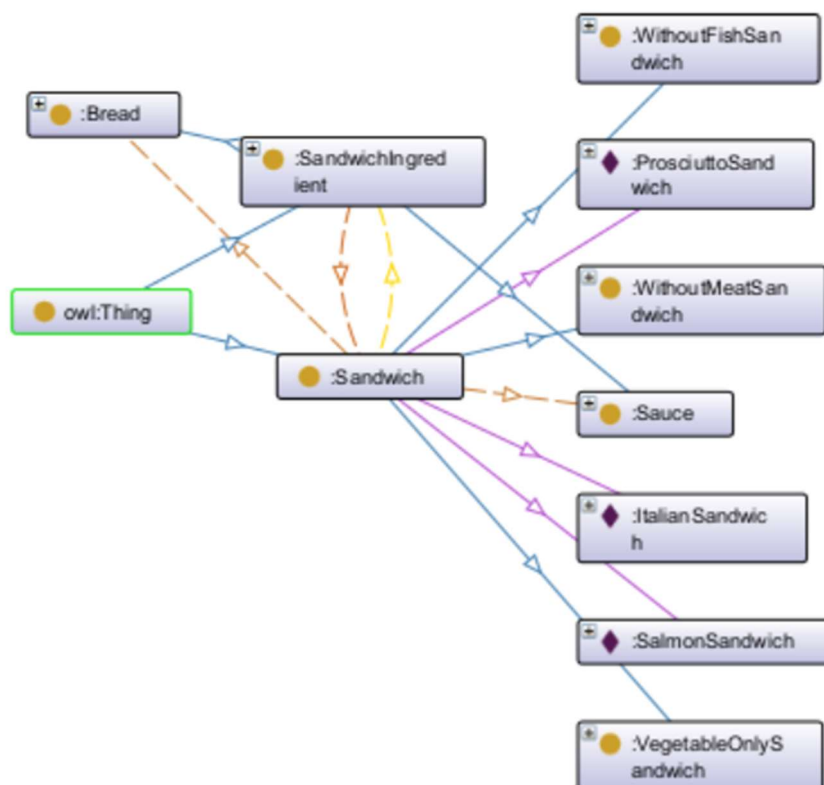
The following pictures show the OntoGraf graph:

The following model shows:

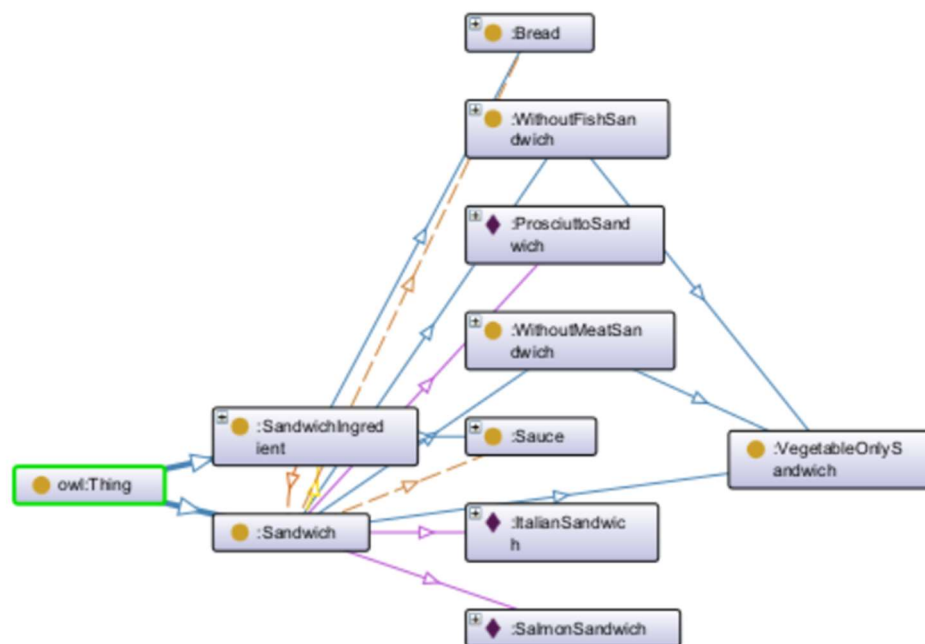
- (1) Classes in our model can be either a type of sandwich or sandwich ingredient.
- (4) The role `hasIngredient` (object property) links elements of the sandwich class and elements of the sandwich ingredient class.
- (11) The role `isIngredientOf` acts as an inverse role of `hasIngredient`.



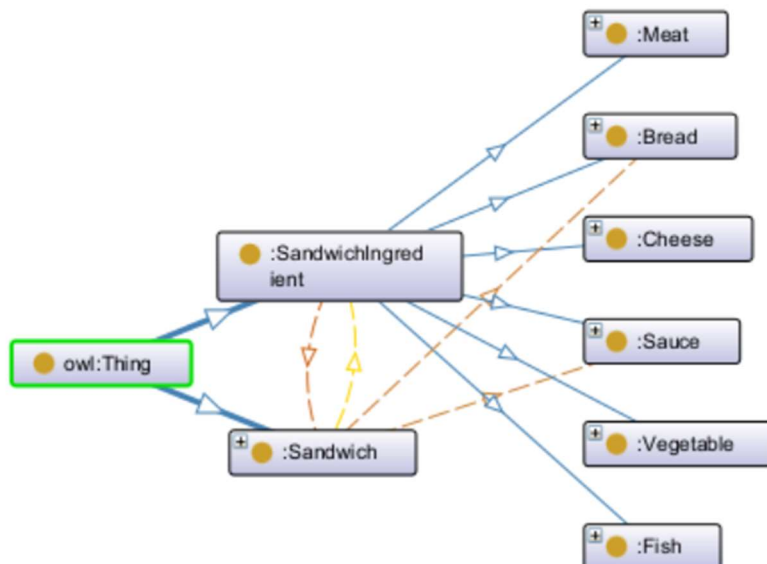
(2.1) Sandwiches are separated in categories by their ingredients: only vegetables, without meat, and without fish.



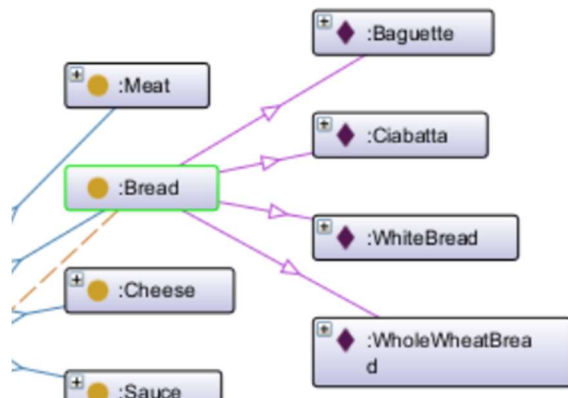
(2.2) Sandwiches that have only vegetables also belong to the categories without meat and without fish.



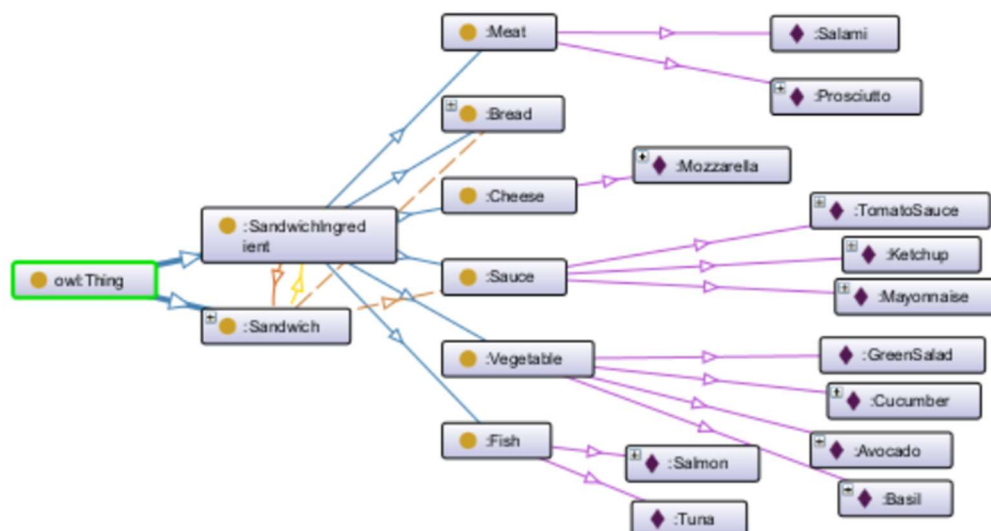
(3.1) Sandwich ingredients can be divided into bread, sauce, vegetable, fish, cheese, and meat with disjoint sets of instances.



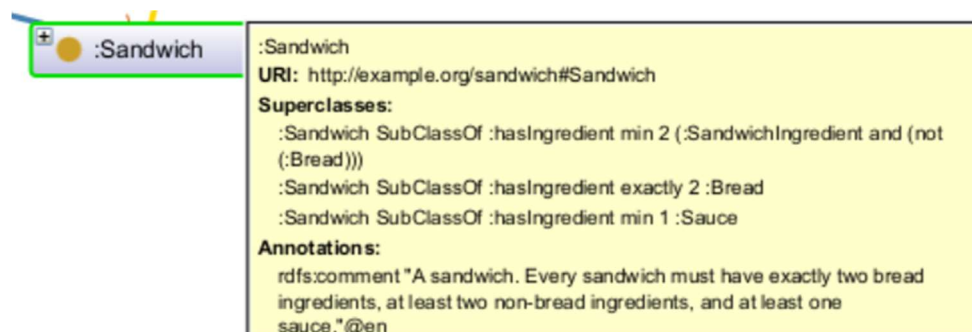
(3.2) Besides different sorts of bread (modelling of two kinds is required)



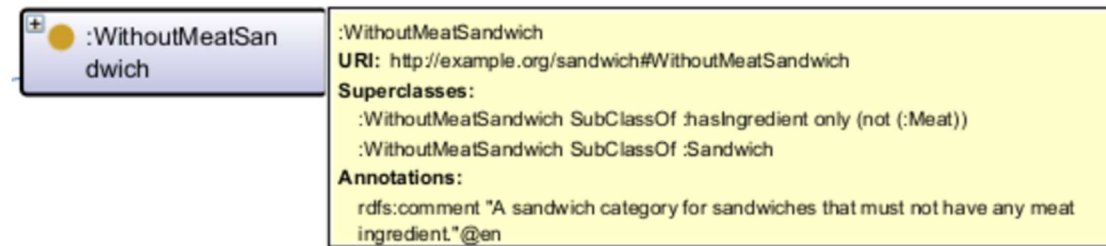
(3.3) typical sandwich ingredients are, for instance, ketchup, cucumber, mayonnaise, green salad, prosciutto (ham), salami, avocado, mozzarella, salmon, and tuna: (in the following are more ingredients as the individual sandwiches are already included, which extend the model)



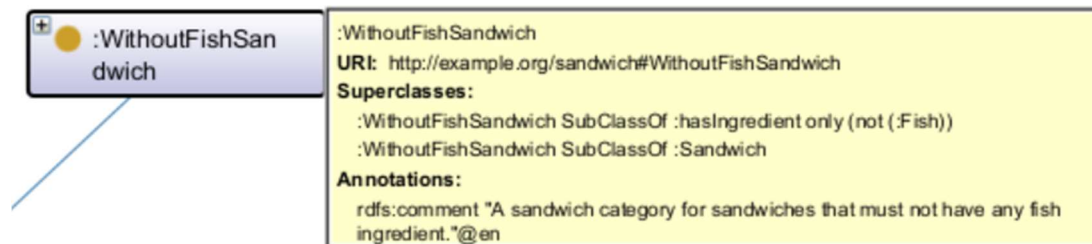
(5) Each sandwich has exactly two pieces of bread and at least two other ingredients where at least one of them is a sauce.



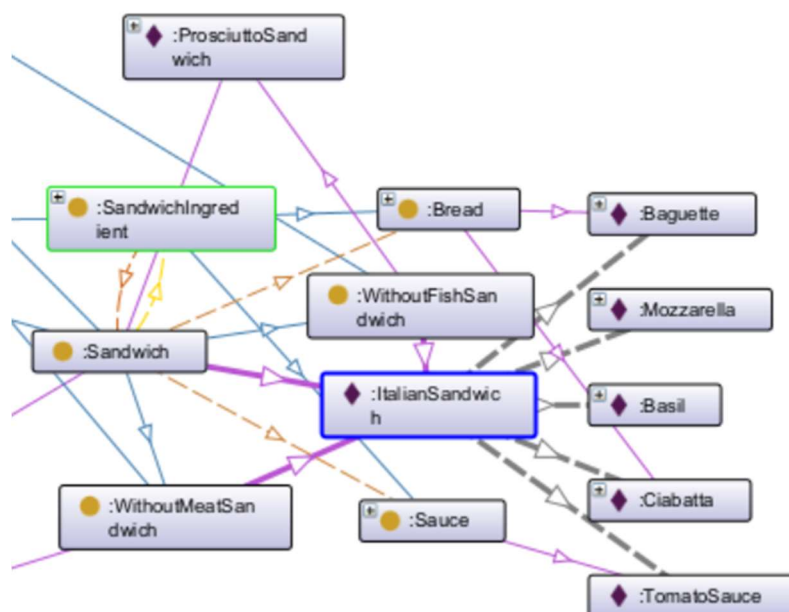
(6) Each sandwich from the category (i.e., class) without meat is not allowed to have any kind of meat as ingredient.



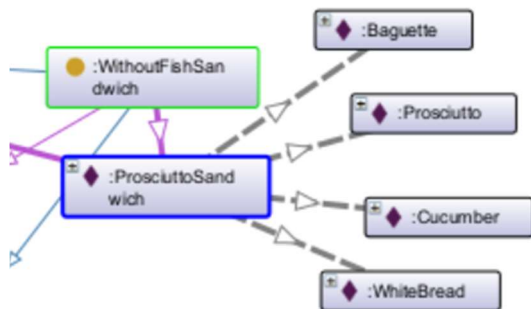
(7) Each sandwich from the category (i.e., class) without fish is not allowed to have any kind of fish as ingredient.



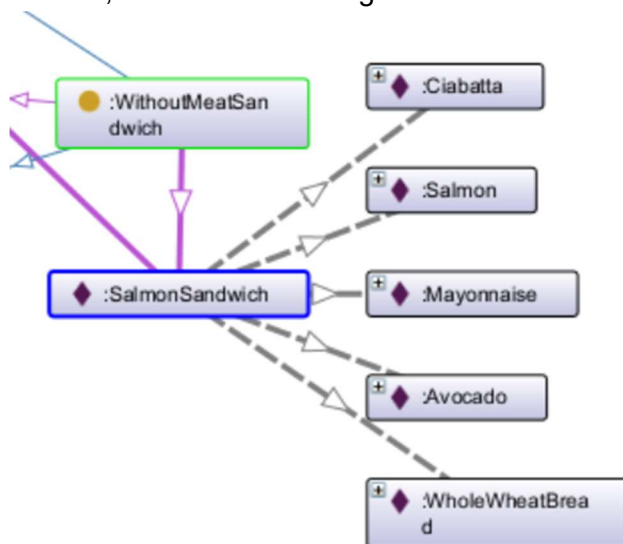
(8) The individual Italian Sandwich has ciabatta, baguette, tomato sauce, basil, and mozzarella as ingredients and is an instance of the classes sandwich without fish and sandwich without meat.



(9) The individual Prosciutto Sandwich has white bread, baguette, ketchup, prosciutto, and cucumber as ingredients and is an instance of the class sandwich without fish.



(10) The individual Salmon Sandwich has whole-wheat bread, ciabatta, mayonnaise, salmon, and avocado as ingredients and is an instance of the class sandwich without meat.



(12) Examine the hasIngredient role with regard to its characteristics and select the appropriate ones. You must explain the choice/decision for all characteristics by adding annotations.

Explanation for (12)

Functional: No. A sandwich has many ingredients (two breads, a sauce, more), not just one.

Inverse functional: No. The same sauce can be used in many different sandwiches.

Transitive: No. We count ingredients (e.g. "exactly 2 bread"), and a property used for counting cannot be transitive.

Symmetric: No. Bread is an ingredient of the sandwich, but the sandwich is not an ingredient of the bread.

Asymmetric: Yes. If bread is an ingredient of a sandwich, the sandwich is never an ingredient of the bread.

Reflexive: No. A sandwich is not an ingredient of itself.

Irreflexive: Yes. A sandwich (or its bread) can never be an ingredient of itself.

Annotations Usage

Annotations: :hasIngredient

Annotations +

rdfs:comment [language: en]

hasIngredient links sandwiches to their ingredients. It is not functional because a sandwich has several ingredients. It is not inverse-functional because the same ingredient can occur in several sandwiches. It is not transitive because an ingredient of an ingredient should not automatically become an ingredient of the sandwich. It is not symmetric because if a sandwich has an ingredient, the ingredient does not have the sandwich as ingredient. Therefore the appropriate characteristics selected here are asymmetric and irreflexive.

Characterist Description: :hasIngredient

☐ Functional
☐ Inverse functional
☐ Transitive
☐ Symmetric
☒ Asymmetric
☐ Reflexive
☒ Irreflexive

Inverse Of +

:isIngredientOf

Domains (intersection) +

:Sandwich

Ranges (intersection) +

:SandwichIngredient

Disjoint With +

SuperProperty Of (Chain) +

The following picture shows the inferred class hierarchy from the reasoner HermiT 1.4.3.456:

